

Week 5 Report

Classification Models — Titanic Survival Prediction

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1 Introduction

This project focused on solving a binary classification problem using the Titanic Survival dataset. Multiple machine learning classification algorithms were trained and compared to predict whether a passenger survived or not. The project emphasized preprocessing, feature engineering, hyperparameter tuning, model evaluation, cross-validation, ROC analysis, confusion matrices, and deployment readiness.

The following models were implemented:

- Logistic Regression
- K-Nearest Neighbours (KNN)
- Decision Tree
- Random Forest

The objective was to identify the best-performing classifier using metrics such as Accuracy, Precision, Recall, F1-Score, and ROC-AUC.

2 Dataset Overview

The Titanic dataset contains passenger information such as age, gender, ticket class, fare, and family relations. The target variable was:

- **Survived**

Key preprocessing steps included:

- Filling missing Age values using median
- Filling missing Embarked values using mode
- Dropping irrelevant columns
- Encoding categorical variables
- Creating FamilySize and IsAlone features
- Applying log transformation to Fare
- Feature scaling using StandardScaler

3 Baseline Logistic Regression

Logistic Regression served as the baseline classifier. Different probability thresholds were analyzed to understand Precision-Recall tradeoffs.

3.1 Important Features

The most influential features included:

- Sex
- Pclass
- Fare
- Age

4 K-Nearest Neighbours

KNN models were trained for values of K from 1 to 30. The best K was selected using F1-score and GridSearchCV.

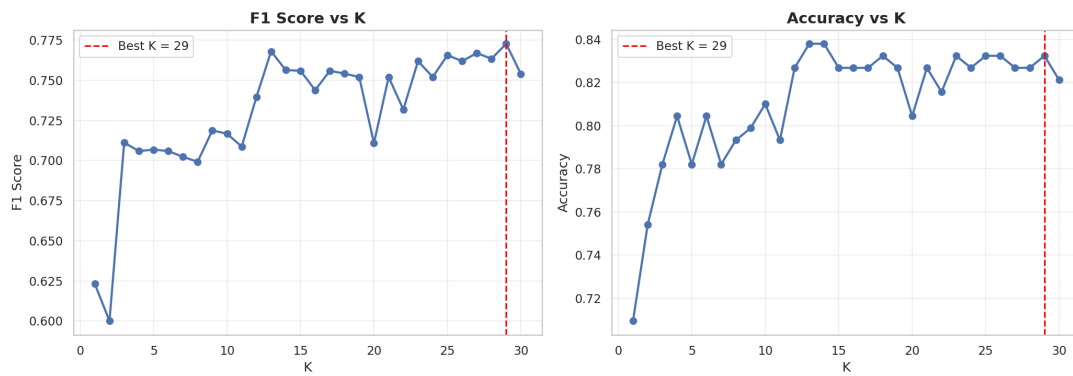


Figure 1: KNN Performance Across Different Values of K

Different distance metrics were also compared:

- Euclidean
- Manhattan
- Minkowski

5 Decision Tree Analysis

Decision Trees were evaluated using different maximum depths to study overfitting behavior.

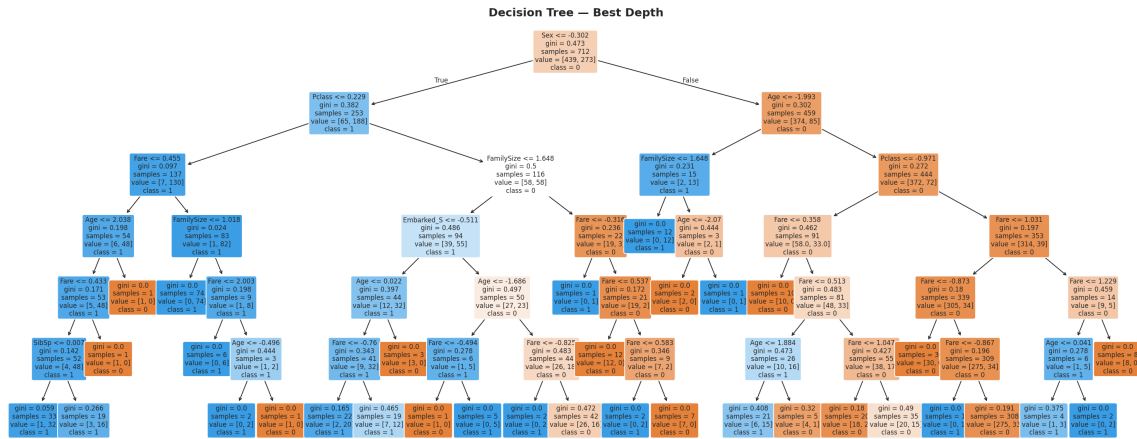


Figure 2: Best Decision Tree Visualization

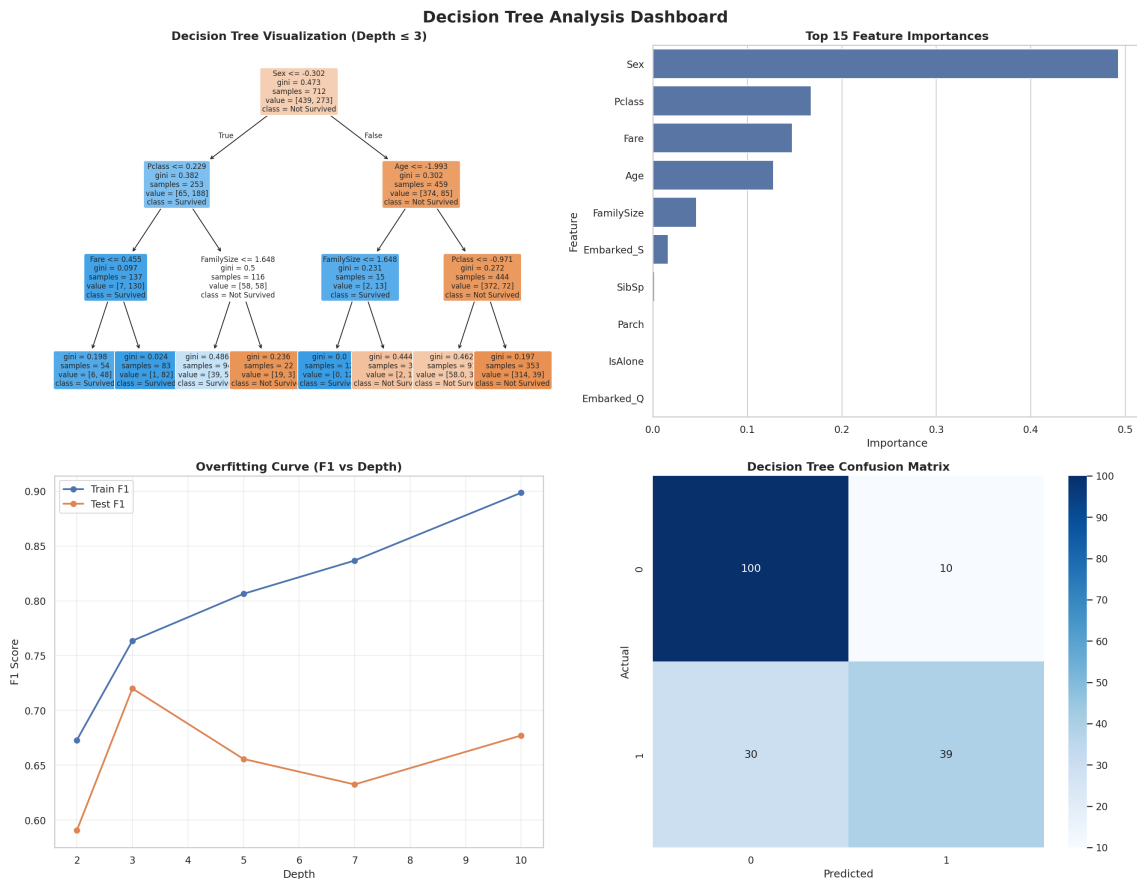


Figure 3: Decision Tree Analysis Dashboard

The first split in the Decision Tree revealed that passenger gender and class were highly predictive of survival.

6 Random Forest Analysis

Random Forest classifiers were trained with different numbers of trees to study performance saturation.

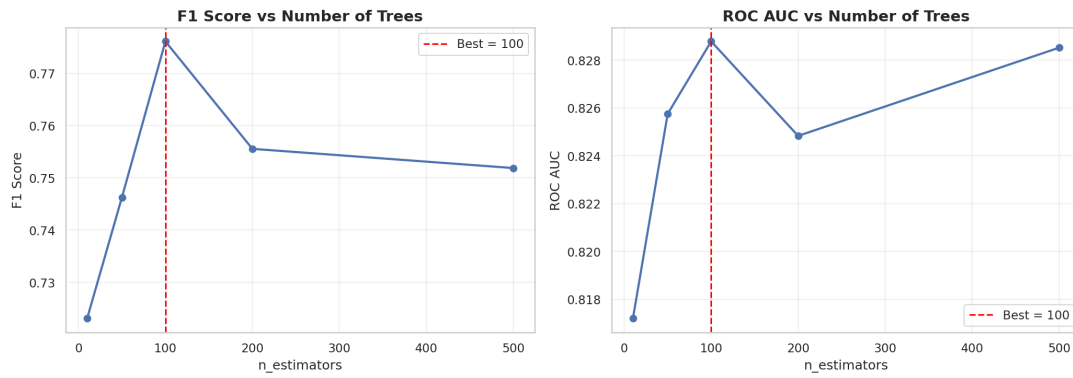


Figure 4: Random Forest Performance vs Number of Trees

Feature importances showed that the strongest survival predictors were:

- Sex
- Fare
- Pclass
- Age

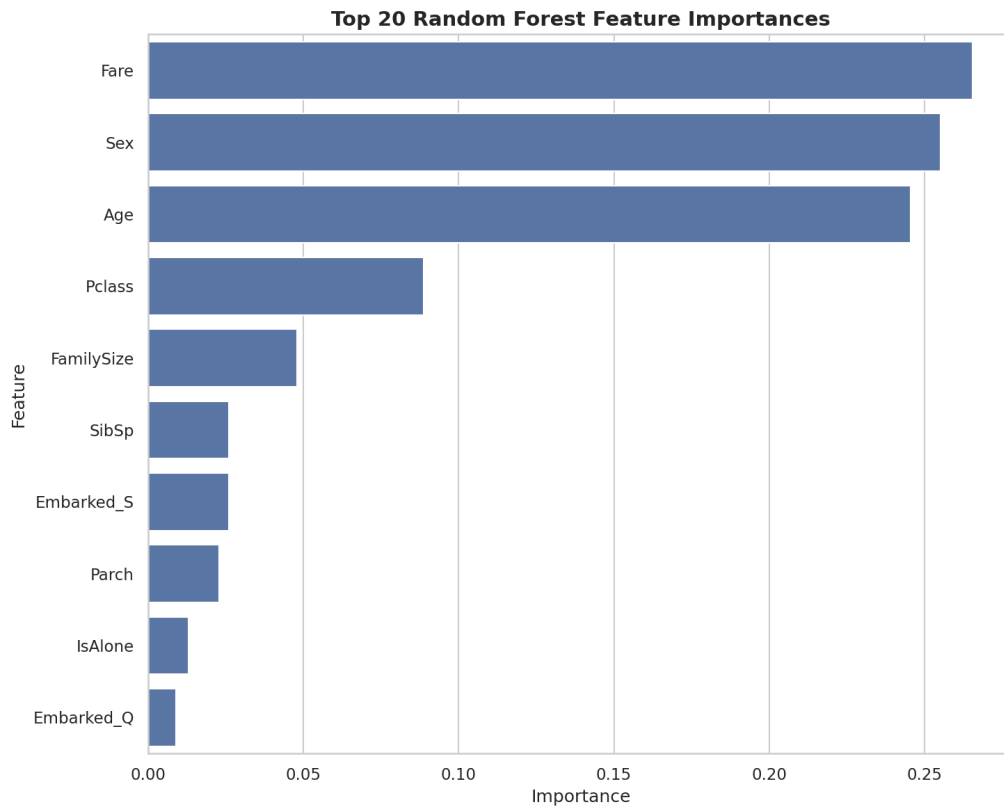


Figure 5: Top Random Forest Feature Importances

7 Final Model Comparison

All classification models were compared using multiple evaluation metrics.

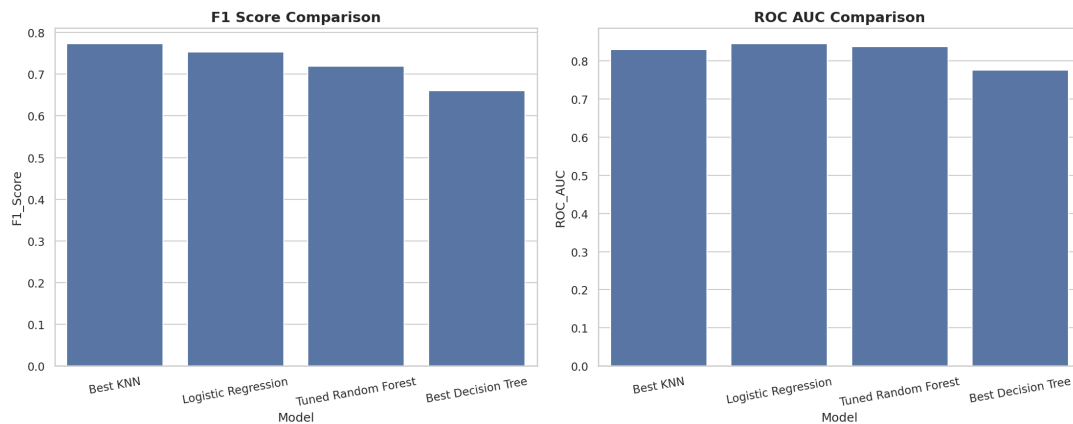


Figure 6: Final Model Comparison

Random Forest achieved the strongest overall performance due to ensemble learning and better nonlinear modeling capability.

8 ROC and Precision-Recall Analysis

ROC and Precision-Recall curves were used for threshold-independent evaluation.

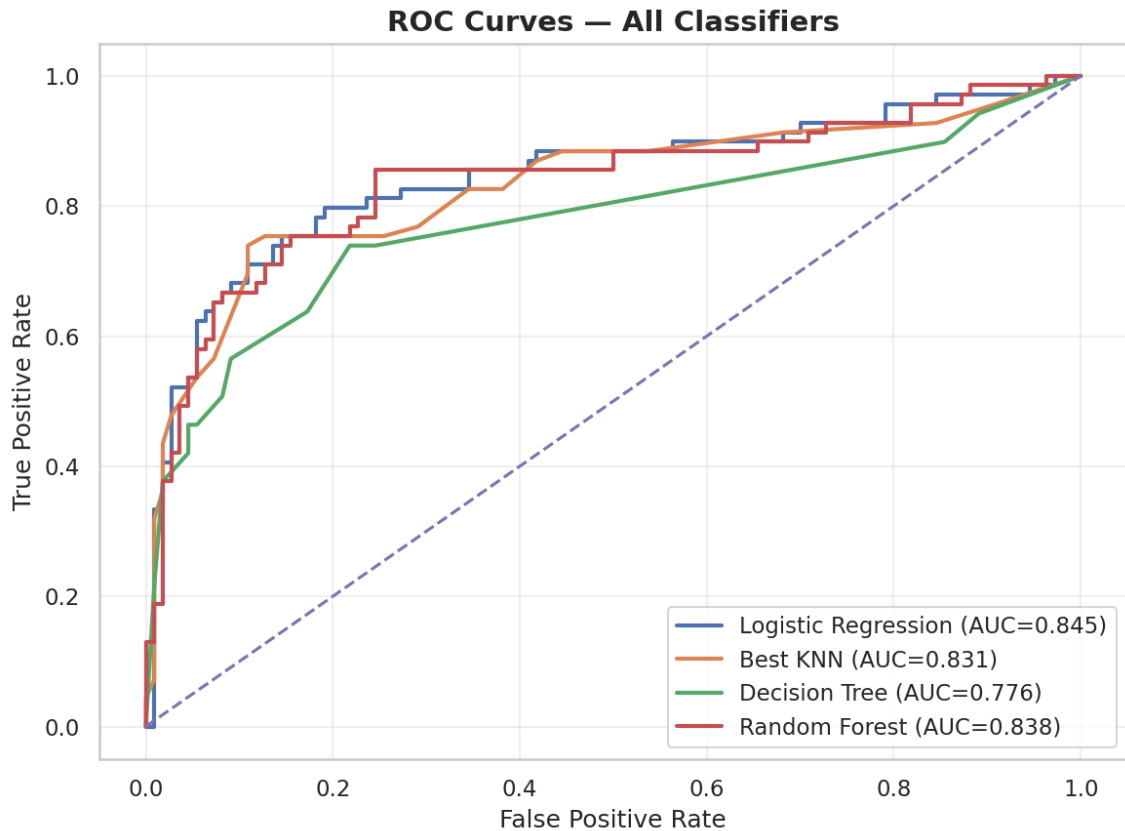


Figure 7: ROC Curves for All Models

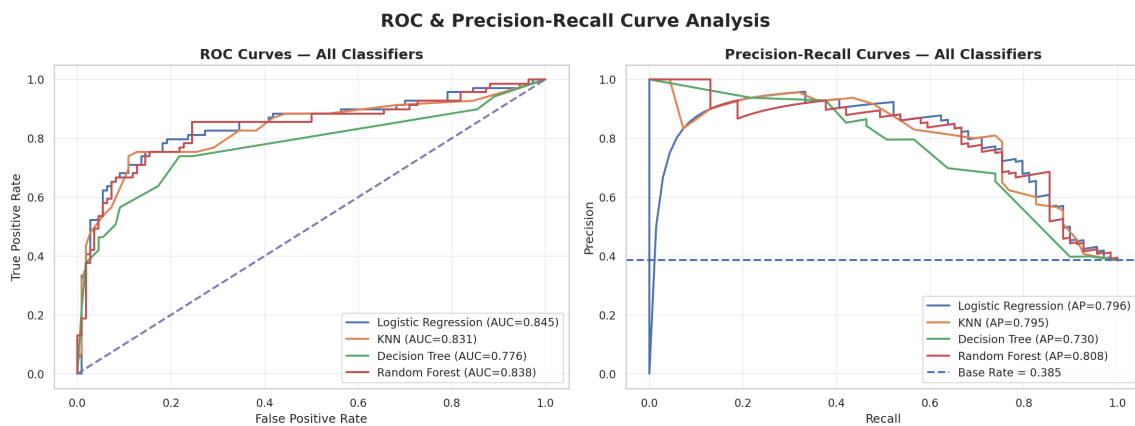


Figure 8: ROC and Precision-Recall Curve Comparison

Precision-Recall curves were particularly useful because classification performance on the positive class was critical.

9 Confusion Matrix Deep Dive

Confusion matrices helped analyze False Positives and False Negatives in detail.

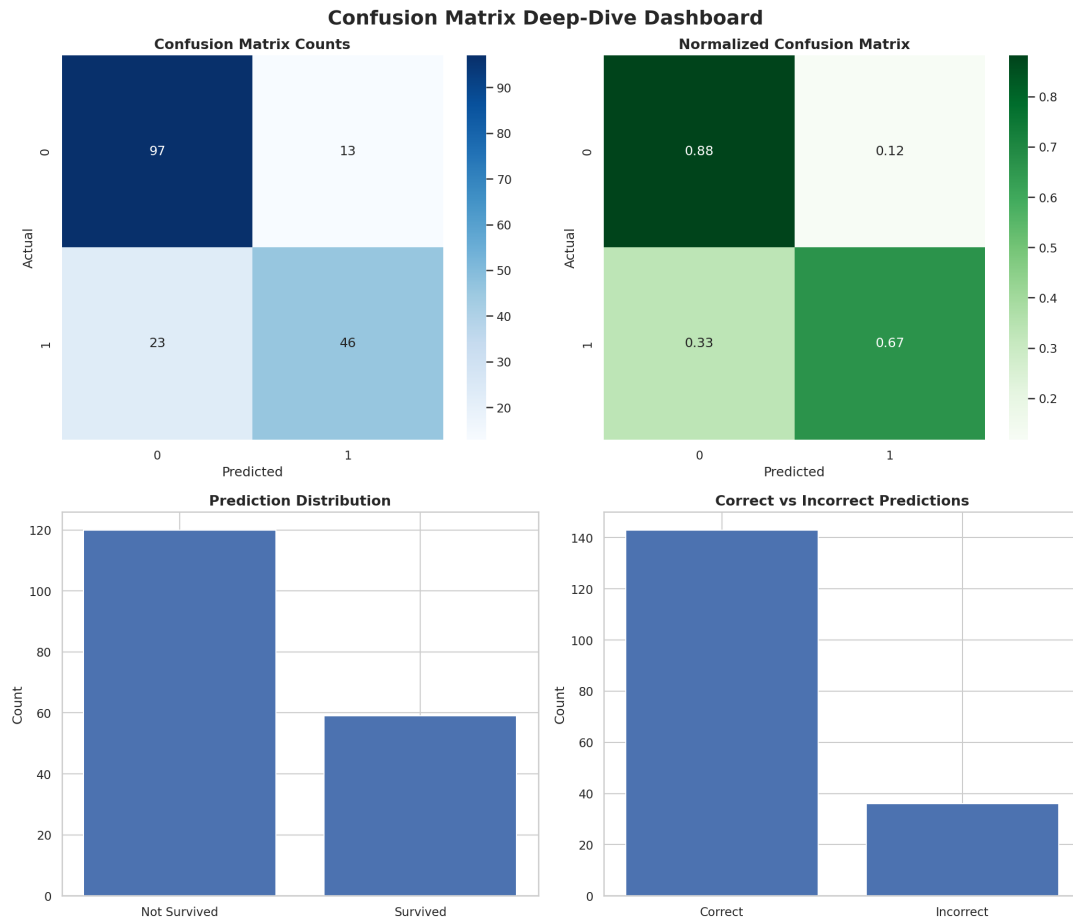


Figure 9: Confusion Matrix Deep-Dive

The analysis revealed that the model performed strongly overall but still struggled with certain edge cases and rare passenger profiles.

10 Cross-Validation Analysis

5-Fold Stratified Cross-Validation was used to estimate real-world model stability.

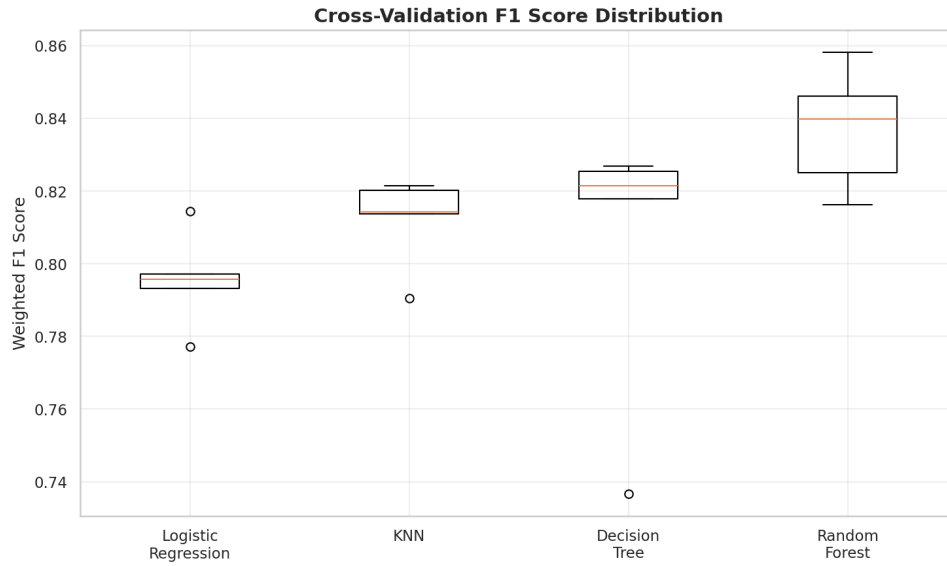


Figure 10: Cross-Validation F1 Score Distribution

Random Forest achieved the highest mean F1 score with strong consistency across folds.

11 Learning Curve Analysis

Learning curves were used to diagnose overfitting and underfitting.

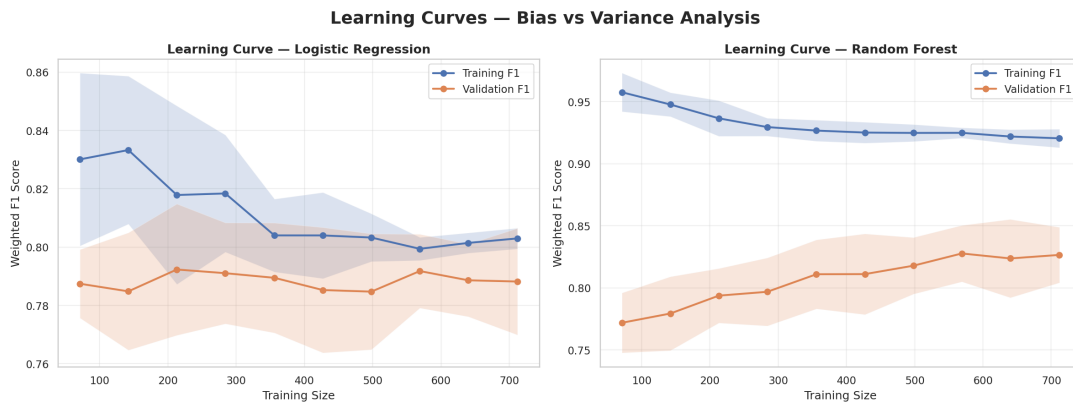


Figure 11: Learning Curves

The curves showed that the Random Forest model generalized well while maintaining strong validation performance.

12 Complete Dashboard

A complete dashboard combining all major evaluation visuals was developed.

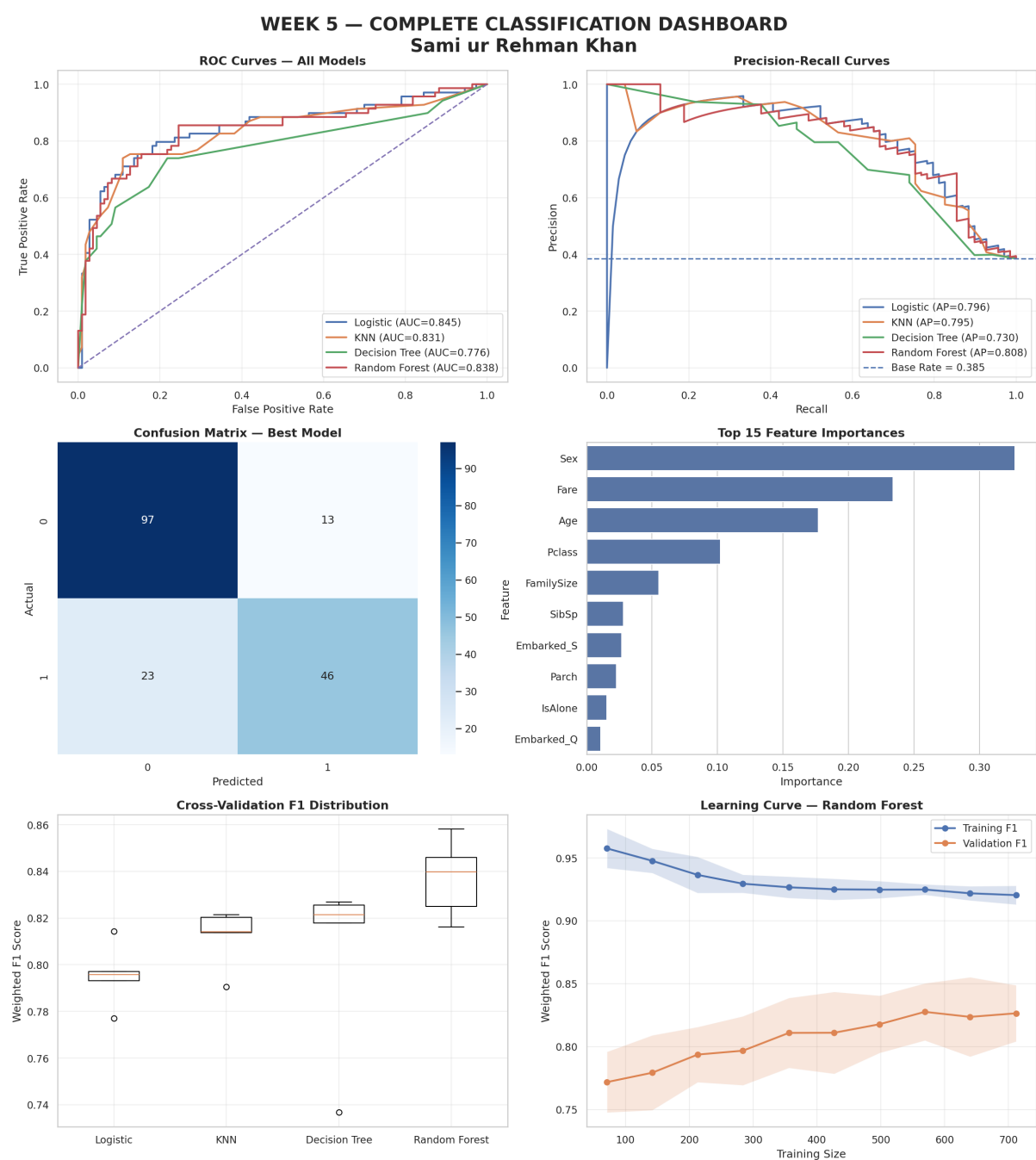


Figure 12: Complete Week 5 Dashboard

13 Prediction Error Analysis

The most confident incorrect predictions were analyzed to understand model weaknesses.

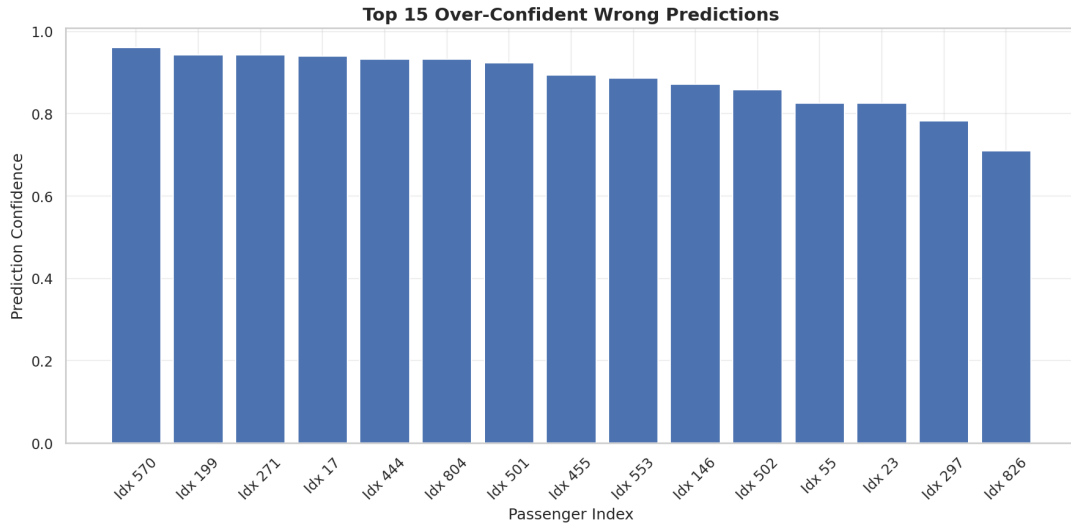


Figure 13: Most Over-Confident Wrong Predictions

These errors often occurred when passengers had unusual combinations of features that contradicted dominant survival patterns.

14 Conclusion

This project demonstrated the complete machine learning workflow for classification problems. Multiple models were trained, evaluated, tuned, and compared using advanced evaluation techniques.

Key achievements included:

- End-to-end classification pipeline development
- Hyperparameter tuning using GridSearchCV
- ROC and Precision-Recall analysis
- Cross-validation and learning curve analysis
- Confusion matrix diagnostics
- Model persistence using joblib
- Professional dashboard generation

Among all models, Random Forest produced the strongest overall performance due to its ensemble structure and ability to model nonlinear relationships effectively.